



MCT344 & MCT342

Industrial Robotics

Sheet (1): Part#1

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Zero Matrix

A matrix in
which every
element is
 0

Denoted as
 0 or O

Size can be
 $m \times n$

Acts as
additive
identity: $A +$
 $0 = A$

Zero Matrix Example

- **Example:**

- $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, Compute $A + 0$.

- **Solution:**

- $A + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

Identity Matrix

Square matrix with 1s on diagonal and 0 elsewhere

Denoted by I

Multiplicative identity:

$$AI = IA = A$$

Identity Matrix Example

- **Example:**

- $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.Compute AI .

- **Solution:**

- $AI = A$;because identity does not change the matrix

Diagonal Matrix

Square matrix with nonzero entries only on main diagonal

Off-diagonal entries are zero

Example:

$$D = \begin{bmatrix} 4 & 0 \\ 0 & 7 \end{bmatrix}$$

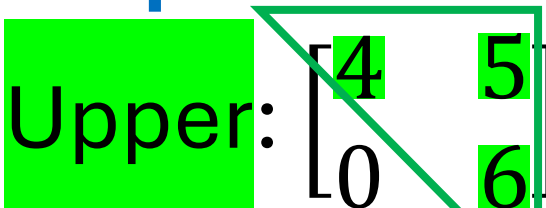
Solution:

Diagonal entries are 4 and 7

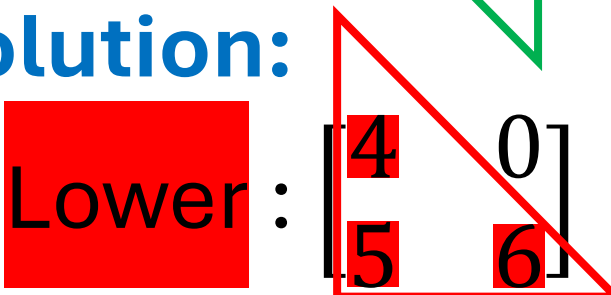
Triangular Matrix

- Upper triangular: zeros below main diagonal
- Lower triangular: zeros above main diagonal

- **Example:**

- Upper: 

- **Solution:**

- Lower: 

Bounded Matrix

A matrix whose
elements are bounded
in magnitude

There exists M such that
 $|a_{ij}| \leq M$ for all entries

Bounded Matrix Example

A matrix is **bounded** if all its entries stay within some fixed number M , meaning:

$$|a_{ij}| \leq M \quad \text{for all elements}$$

Consider the matrix:

$$A = \begin{bmatrix} 1.2 & -3.5 & 0.8 \\ 2.1 & -1.0 & 4.3 \\ -2.7 & 3.9 & -0.6 \end{bmatrix}$$

Step 1: Look at absolute values of all entries

$$|A| = \begin{bmatrix} 1.2 & 3.5 & 0.8 \\ 2.1 & 1.0 & 4.3 \\ 2.7 & 3.9 & 0.6 \end{bmatrix}$$

Step 2: Find the largest value

The largest absolute value is 4.3

Step 3: State the bound

Since all elements satisfy

$$|a_{ij}| \leq 5$$

Matrix **A** is bounded with bound **M = 5**

Matrix Addition

- Defined only for matrices of same size
- Add corresponding elements
- Commutative and associative

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

Solution:

$$A + B = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$$

Matrix Transpose

Rows become columns

$$(A^T)^T = A$$

$$(A + B)^T = A^T + B^T$$

$$(AB)^T = B^T A^T$$

Transpose Example

- **Example:**

- $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$

- **Solution:**

- $A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$

Symmetric Matrices

A matrix A is
symmetric if $A^T = A$

Diagonal entries
can be any value

Skew-Symmetric Matrices

A matrix A is skew-symmetric if $A^T = -A$

Diagonal entries must be 0

Symmetric & Skew Example

- **Example:**

- Symmetric: $\begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$

- Skew: $\begin{bmatrix} 0 & 4 \\ -4 & 0 \end{bmatrix}$

- **Solution:**

- Check if transpose equals:
A or $-A$

Inverse of a Matrix

Only square non-singular matrices have inverses

$$AA^{-1} = A^{-1}A = I$$

Singularity Check

Matrix is singular if determinant = 0

Singular matrices do not have inverses

Example:

- $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, check singularity.

Solution:

- $\det(A) = 1 \cdot 4 - 2 \cdot 2 = 0$ singular $\rightarrow$ no inverse

Singularity Check

Practice Example:

- Check if $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$ is singular.

Answer:

- Rows are linearly dependent
- $\rightarrow$ determinant = 0 $\rightarrow$ Singular

Inner Product of Vectors

Also called dot product

$u^T \cdot v$ = sum of products of components

Produces a scalar

Example:

$$\bullet u = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, v = \begin{bmatrix} 4 \\ 0 \\ -1 \end{bmatrix}$$

Solution:

$$\bullet u^T \cdot v = 1*4 + 2*0 + 3*(-1) = 1$$

Outer Product of Vectors

$u v^T$ produces a matrix

Result is $m \times n$ if u is $m \times 1$ and v is $n \times 1$

Produces a matrix

Example:

- $u = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, v = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$

Solution:

- $u v^T = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$

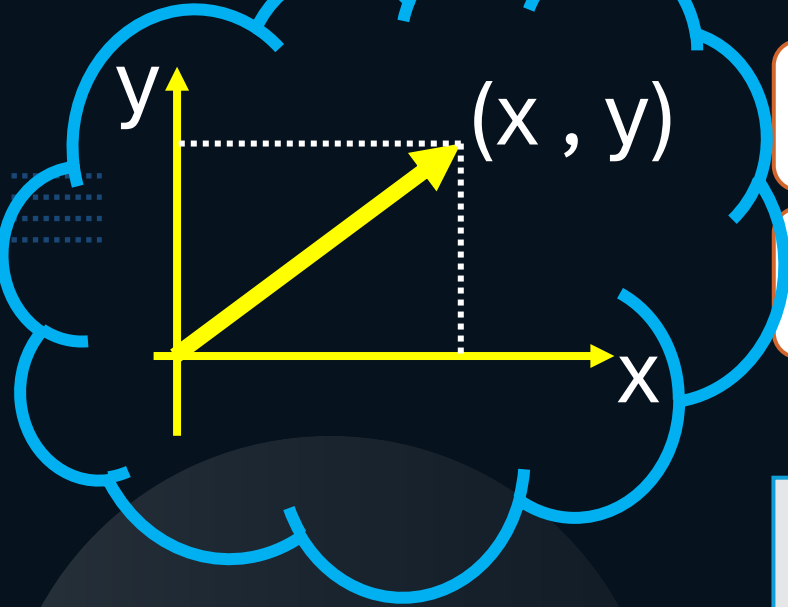
Orthogonal Vectors

- Two vectors are orthogonal if their dot product is zero.
Example: $u^T \cdot v = 0 \rightarrow u \text{ \& } v \text{ are orthogonal.}$

Normalized Vectors

- A vector with norm equal to 1
- Obtained by dividing vector by its norm.

Example: $\|u\| = \sqrt{u_x^2 + u_y^2 + u_z^2} = 1$



Euclidean Norm of a Vector

Length or magnitude of a vector

Euclidean norm: $||v|| = \text{sqrt}(\text{sum of squares})$

Example:

- $v = [3, 4] = 3\hat{i} + 4\hat{j}$

Solution:

- $||v|| = 5 \rightarrow \text{unit vector} = v / ||v|| = [3/5, 4/5]$

Geometric Meaning of Dot Product

$$u \cdot v = ||u|| ||v|| \cos(\theta)$$

If dot product = 0 $\rightarrow$
vectors are perpendicular

Positive $\rightarrow$ angle $< 90^\circ$,
Negative $\rightarrow$ angle $> 90^\circ$

Why Transpose Matters

Switches rows and columns

Used in dot products: $u \cdot v = u^T v$

Helps define orthogonality: $A^T A$

Sheet (1) Problems

Sheet (1) – Pb.1

A mobile robot is at $P_0 = (2, 1)$ m. It moves by displacement $d = (3, -0.5)$ m. **Find the new position P_1 .**

Solution:

Add components: $P_1 = P_0 + d$. So $P_1 = (2+3, 1-0.5) = (5, 0.5)$ m.

Sheet (1) – Pb.2

Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$.

a) Calculate: $A \times B$ and $B \times A$

Solution:

$$A \times B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 2 \\ 3 & 1 \end{bmatrix}$$

$$B \times A = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$$

$A \times B \neq B \times A$ (Not commutative)

Sheet (1) – Pb.2

Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$.

b) Calculate: $B^{-1}.A^{-1}$

Solution:

$$A \times B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 2 \\ 3 & 1 \end{bmatrix}$$

$$B^{-1}.A^{-1} = (A.B)^{-1} = \frac{1}{7-6} \begin{bmatrix} 1 & -2 \\ -3 & 7 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -3 & 7 \end{bmatrix}$$

Sheet (1) – Pb.2

Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$.

c) Calculate: $A^T \cdot B^T$

Solution:

$$B \times A = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$$

$$A^T \cdot B^T = (B \cdot A)^T = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$$

Sheet (1) – Pb.3

$$\text{Let } A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \text{ and } B = \begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}, \text{ calculate: } A*B$$

Solution:

$$A \times B = \begin{bmatrix} (1)(9)+(2)(6)+(3)(3)=9+12+9=30 & (1)(8)+(2)(5)+(3)(2)=8+10+6=24 & (1)(7)+(2)(4)+(3)(1)=7+8+3=18 \\ (4)(9)+(5)(6)+(6)(3)=36+30+18=84 & (4)(8)+(5)(5)+(6)(2)=32+25+12=69 & (4)(7)+(5)(4)+(6)(1)=28+20+6=54 \\ (7)(9)+(8)(6)+(9)(3)=63+48+27=138 & (7)(8)+(8)(5)+(9)(2)=56+40+18=114 & (7)(7)+(8)(4)+(9)(1)=49+32+9=90 \end{bmatrix}$$

$$= \begin{bmatrix} 30 & 24 & 18 \\ 84 & 69 & 54 \\ 138 & 114 & 90 \end{bmatrix}$$

Sheet (1) – Pb.4

If $u = (4, 0)$ and $v = (0, 3)$. Compute the dot product of the two vectors ($u \cdot v$) then decide if u and v are orthogonal/perpendicular.

Solution:

$u \cdot v = 4*0 + 0*3 = 0$. If $u \cdot v = |u| \cdot |v| \cdot \cos(\theta) = 0$, then the angle is 90 degrees, so u and v are **orthogonal/perpendicular**.

Sheet (1) – Pb.5

A link points from $A = (1, 2)$ to $B = (5, 5)$. **Find the unit vector (direction) from A to B.**

Solution:

Compute the displacement: $d = B - A = (4, 3)$.

Compute its length: $|d| = \sqrt{4^2 + 3^2} = 5$.

Unit vector: $\hat{d} = \frac{d}{|d|} = \left(\frac{4}{5}, \frac{3}{5}\right) = (0.8, 0.6)$.

Sheet (1) – Pb.6

If $A = (1, 3, 4)$ and $B = (2, 7, -5)$. Find the cross product ($A \times B$) then calculate the angle between A and B .

Solution:

$$A \times B = \begin{vmatrix} i & j & k \\ 1 & 3 & 4 \\ 2 & 7 & -5 \end{vmatrix} = \begin{vmatrix} 3 & 4 \\ 7 & -5 \end{vmatrix} i - \begin{vmatrix} 1 & 4 \\ 2 & -5 \end{vmatrix} j + \begin{vmatrix} 1 & 3 \\ 2 & 7 \end{vmatrix} k = -43i + 13j + k$$

$$|A| = \sqrt{1^2 + 3^2 + 4^2} = \sqrt{26}$$

$$|B| = \sqrt{2^2 + 7^2 + (-5)^2} = \sqrt{78}$$

$$\sin(\theta) = \frac{|A \times B|}{|A| \cdot |B|} = \frac{\sqrt{-43^2 + 13^2 + 1^2}}{\sqrt{26} \cdot \sqrt{78}} = \frac{\sqrt{-43^2 + 13^2 + 1^2}}{\sqrt{26} \cdot \sqrt{78}} = 0.998$$

$$\therefore \theta = \sin^{-1} 0.998 = 1.5 \text{ rad} = 1.5 \text{ rad} * \frac{180}{\pi} \text{ degree} = 86.18^\circ$$

Sheet (1) – Pb.7

Calculate the rotation matrix (R) to rotate the blue arrow to the green arrow by an angle **theta** as shown below in **Fig.1**.

Solution:

$$x_1 = r \cdot \cos(\varphi) , y_1 = r \cdot \sin(\varphi)$$

$$x_2 = r \cdot \cos(\varphi + \theta) , y_2 = r \cdot \sin(\varphi + \theta)$$

$$x_2 = r \cdot \cos(\varphi + \theta)$$

$$= r[\cos(\varphi) \cos(\theta) - \sin(\varphi) \sin(\theta)] = [x_1 \cos(\theta) - y_1 \sin(\theta)]$$

$$y_2 = r \cdot \sin(\varphi + \theta)$$

$$= r[\sin(\varphi) \cos(\theta) + \cos(\varphi) \sin(\theta)] = [y_1 \cos(\theta) + x_1 \sin(\theta)]$$

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$$
$$\therefore R = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

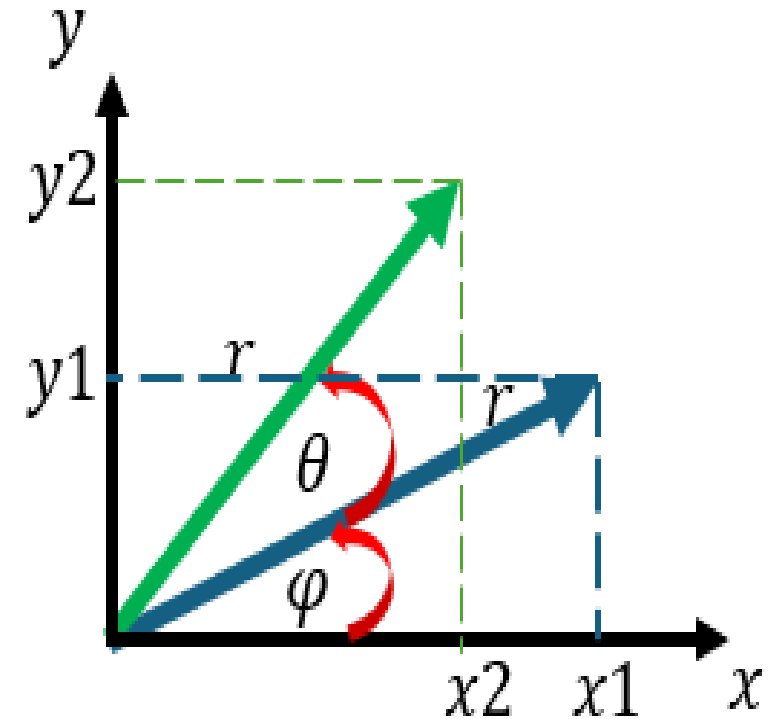


Fig.1

Sheet (1) – Pb.8

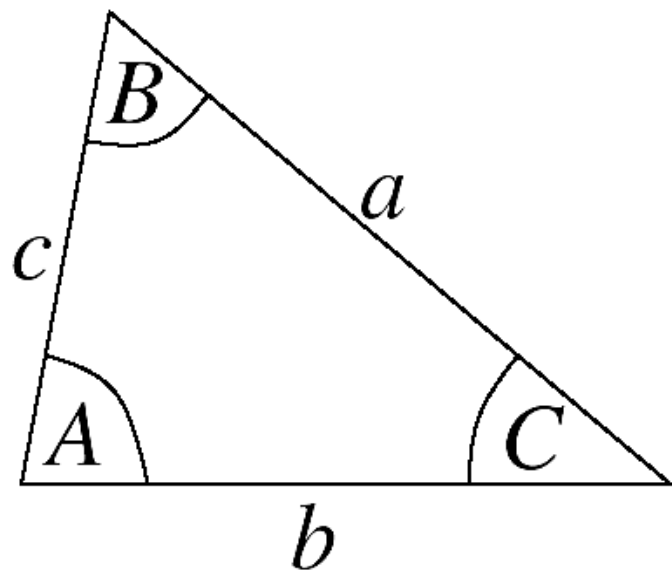
Find the total rotation matrix of a joint that first rotates by 45 degrees then by 30 degrees.

Solution:

Applying to a vector $\mathbf{v}$: first $\mathbf{v}_1 = \mathbf{Rotation45} * \mathbf{v}$, then $\mathbf{v}_2 = \mathbf{Rotation30} * \mathbf{v}_1$.
So, $\mathbf{v}_2 = (\mathbf{Rotation30} * \mathbf{Rotation45}) * \mathbf{v}$ and the combined matrix is $\mathbf{R}_{total} = \mathbf{Rotation30} * \mathbf{Rotation45}$.

In 2D, rotations add: $30 + 45 = 75$ degrees, so $\mathbf{R}_{total} = \mathbf{Rotation75}$.

Remember



Sine Rule

$$\frac{a}{\sin(A)} = \frac{b}{\sin(B)} = \frac{c}{\sin(C)}$$

(for finding sides)

or $\frac{\sin(A)}{a} = \frac{\sin(B)}{b} = \frac{\sin(C)}{c}$

(for finding angles)

Cosine Rule

$$a^2 = b^2 + c^2 - 2bc \cos(A)$$

(for finding sides)

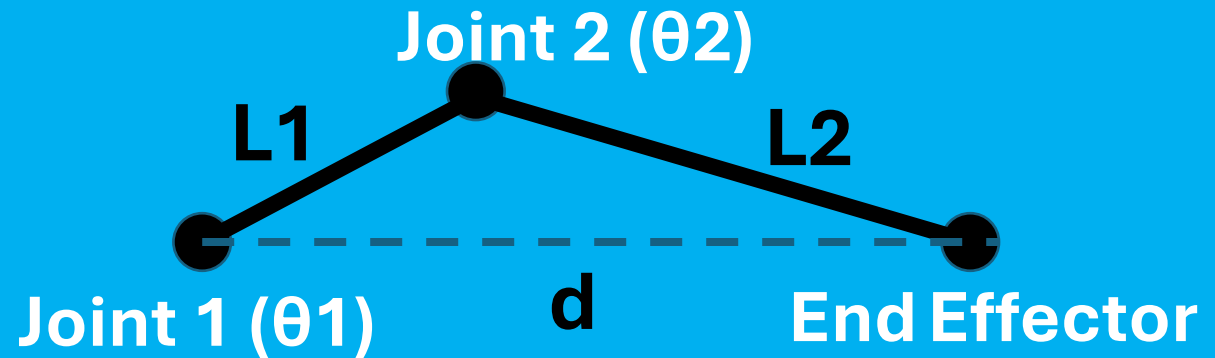
or $\cos(A) = \frac{b^2 + c^2 - a^2}{2bc}$

(for finding angles)

Workspace Limits

Maximum reach = $L1 + L2$

Minimum reach = $|L1 - L2|$



Sheet (1) – Pb.9

A planar 2-link (2R) robot has link lengths $L1 = 4$ cm and $L2 = 3$ cm. The end-effector target point is at distance $d = 5$ cm horizontally from the base ($x = 0, y = 0$), target lies in ($x = 5, y = 0$). Find the elbow angle (the included angle between the two links).

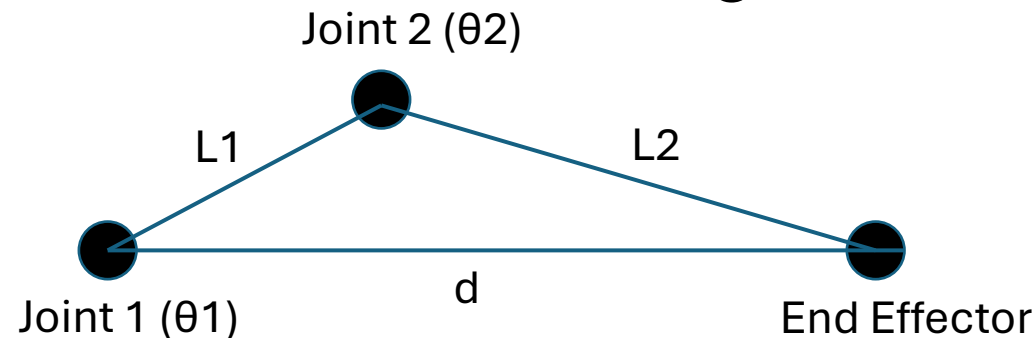
Solution:

Form a triangle with sides **L1**, **L2**, and **d** (base-to-target).

Apply the cosine rule: $d^2 = L1^2 + L2^2 - 2 * L1 * L2 * \cos(\theta_2)$.

Substitute: $5^2 = 4^2 + 3^2 - 2 * 4 * 3 * \cos(\theta_2) \rightarrow 25 = 25 - 24 * \cos(\theta_2)$.

So, $\cos(\theta_2) = 0$ and $\theta_2 = 90$ degrees.



Sheet (1) – Pb.10

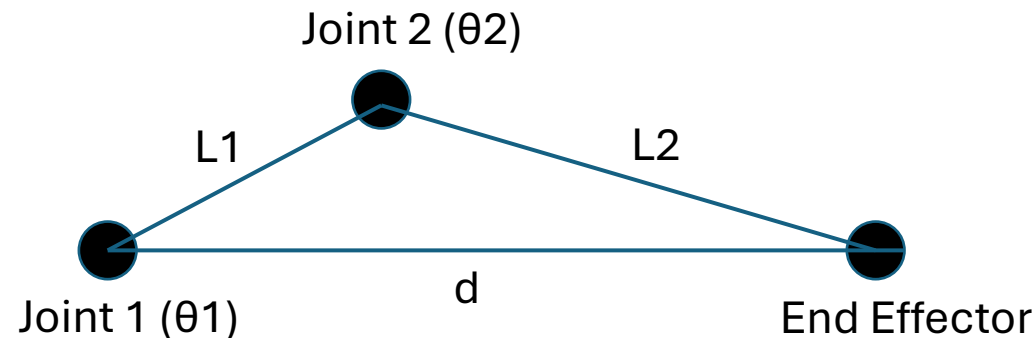
A 2R robot has $L1 = 6$ cm, $L2 = 4$ cm. The target distance is $d = 11$ cm horizontally from the base ($x = 0, y = 0$), target lies in ($x = 11, y = 0$). Decide if the point is reachable and how many inverse kinematic solutions exist (0/1/2).

Solution:

Outer reach = $L1 + L2 = 10$ cm.

Since, $d = 11$ cm $>$ 10 cm, the target is outside the workspace.

Therefore, there are **0 inverse-kinematics solutions (unreachable)**.



Sheet (1) – Pb.11

Consider the triangle in **Fig.2** where $a = 5$ cm, $b = 7$ cm, and angle $A = 30$ degrees. Use the sine rule to determine how many possible triangles exist, and find possible angles B .

Solution:

Using sine rule: $\frac{\sin(A)}{a} = \frac{\sin(B)}{b} \rightarrow \sin(B) = b \cdot \frac{\sin(A)}{a}$

$$\therefore \sin(B) = 7 \cdot \sin(30) / 5 = 0.7.$$

So, B can be $B_1 = \sin^{-1} 0.7 \approx 44.4$ degrees OR $B_2 = 180 - 44.4 = 135.6$ degrees.

Both are valid if $A + B < 180$, which holds for both cases ($30 + 44.4$ and $30 + 135.6$).

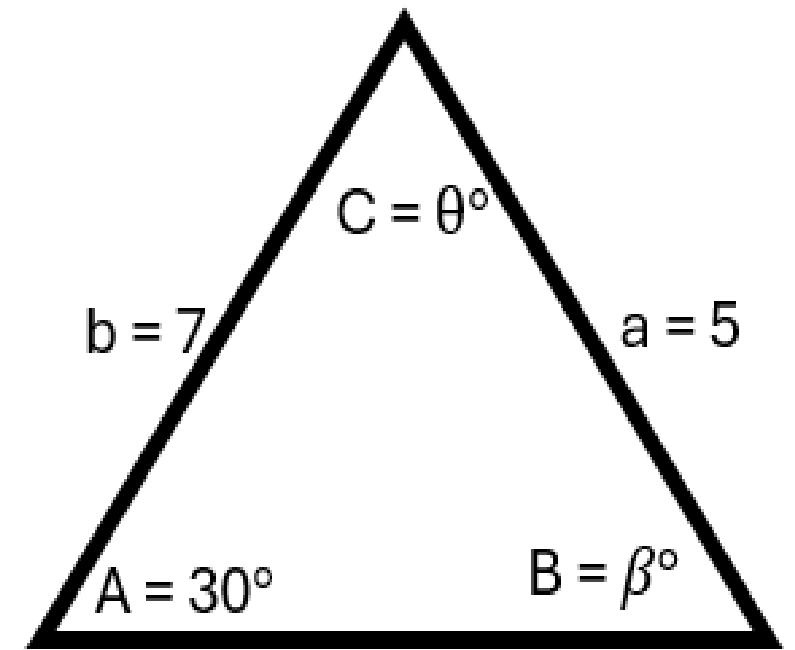


Fig.2

Sheet (1) – Pb.23

A 2R planar arm has $L1 = 4$ cm, $L2 = 3$ cm. Joint angles are $\theta_1 = 30$ degrees and $\theta_2 = 45$ degrees (θ_2 relative to link-1). Compute the end-effector position (x, y).

Solution:

$\theta_{12} = \theta_1 + \theta_2 = 30 + 45 = 75$ degrees.

$x = L1 * \cos(\theta_1) + L2 * \cos(\theta_{12})$,

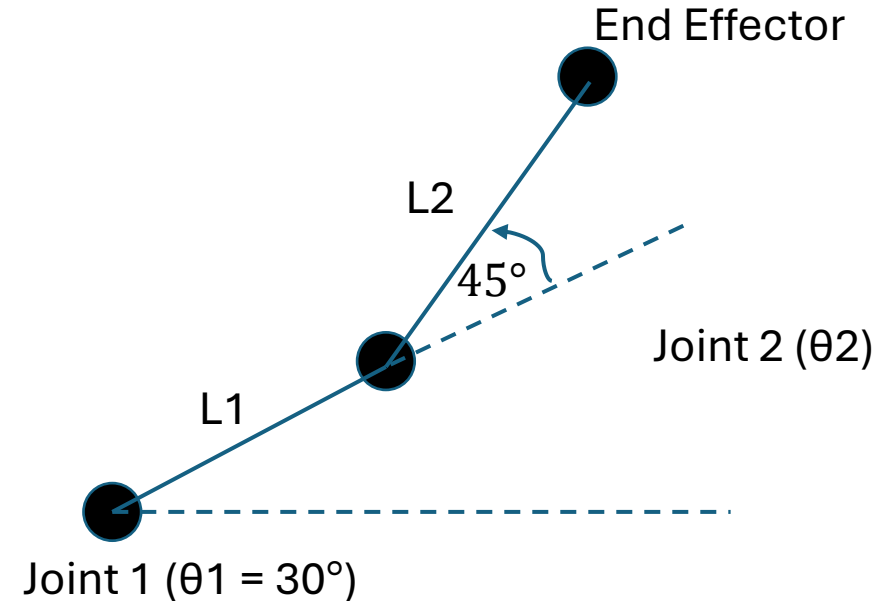
$y = L1 * \sin(\theta_1) + L2 * \sin(\theta_{12})$.

$\cos 30 = 0.866$, $\sin 30 = 0.5$

$\cos 75 = 0.259$, $\sin 75 = 0.966$.

$x = 4 * 0.866 + 3 * 0.259 = 3.464 + 0.777 = 4.241$ cm.

$y = 4 * 0.5 + 3 * 0.966 = 2.000 + 2.898 = 4.898$ cm.



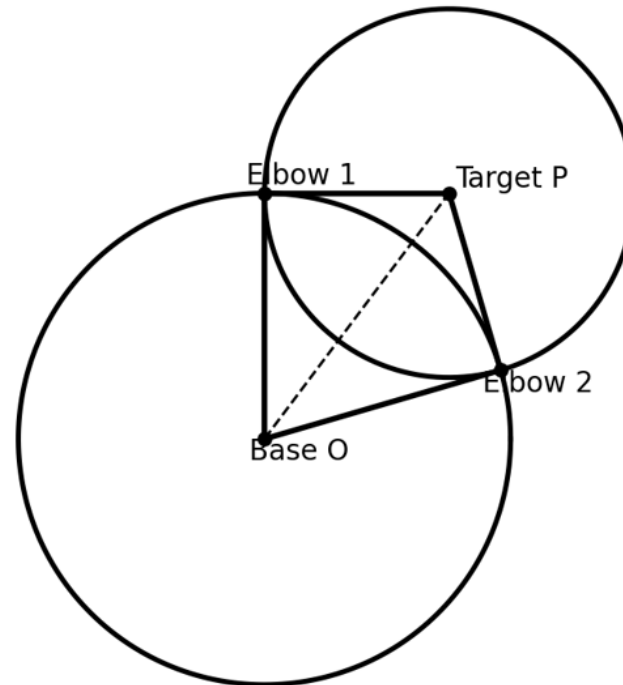
Sheet (1) – Pb.24

A 2R arm has $L1 = 4$ cm, $L2 = 3$ cm. Target is $P = (3,4)$ cm.

- a) Using compass: draw a circle centered at O with radius $L1$ and a circle centered at P with a radius $L2$.
- b) How many intersections are there? What does that mean for IK solutions?
- c) Use cosine rule to compute the elbow angle θ_2 .

Solution:

- a) Draw $\text{circle}(O, L1)$ and $\text{circle}(P, L2)$.



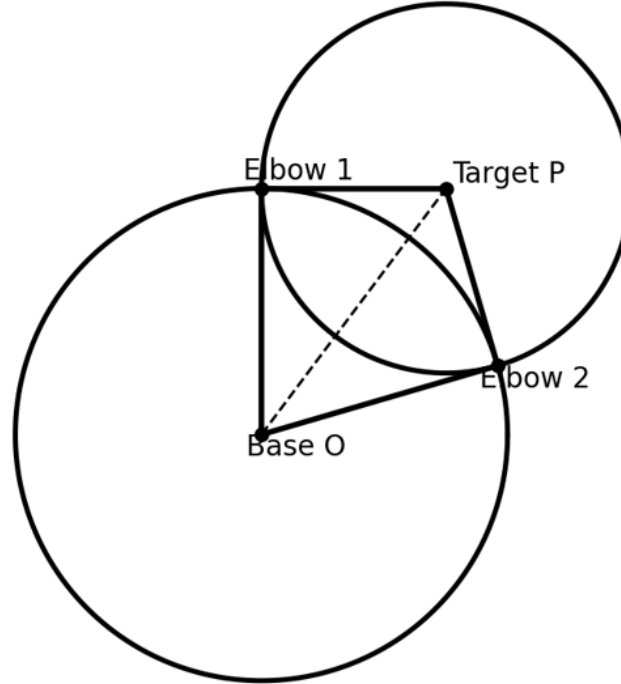
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- a) Using compass: draw a circle centered at O with radius $L1$ and a circle centered at P with a radius $L2$.
- b) How many intersections are there? What does that mean for IK solutions?
- c) Use cosine rule to compute the elbow angle θ_2 .

Solution:

- b) 2 intersections are existing.
therefore, there are two IK solutions.
(elbow up and elbow down)



Sheet (1) – Pb.24

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- a) Using compass: draw a circle centered at O with radius $L1$ and a circle centered at P with a radius $L2$.
- b) How many intersections are there? What does that mean for IK solutions?
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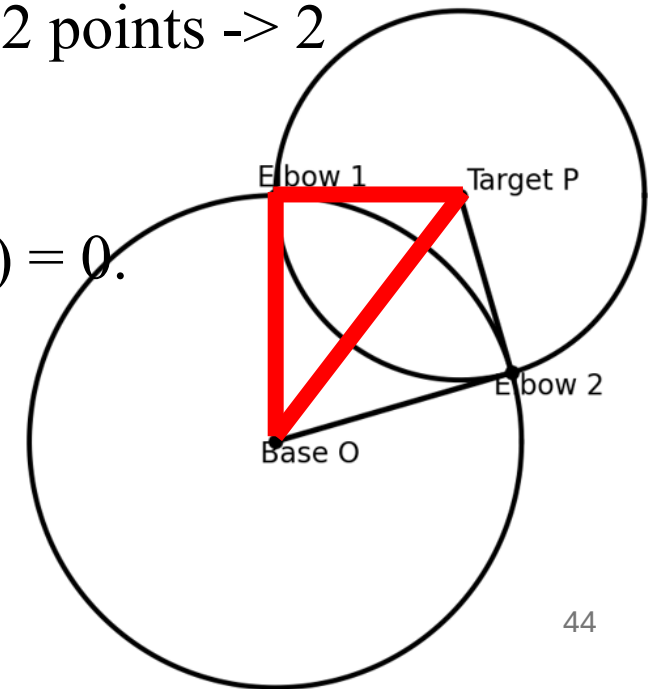
Solution:

- c) For these values, distance $OP = 5$ and the circles intersect in 2 points $\rightarrow 2$ solutions.

Using cosine rule on triangle (O, elbow-up, $OP=5$):

$$\cos(\theta_2) = (L1^2 + L2^2 - d^2) / (2 * L1 * L2) = (16 + 9 - 25) / (24) = 0.$$

So, $\theta_2 = 90$ degrees.





Any questions?